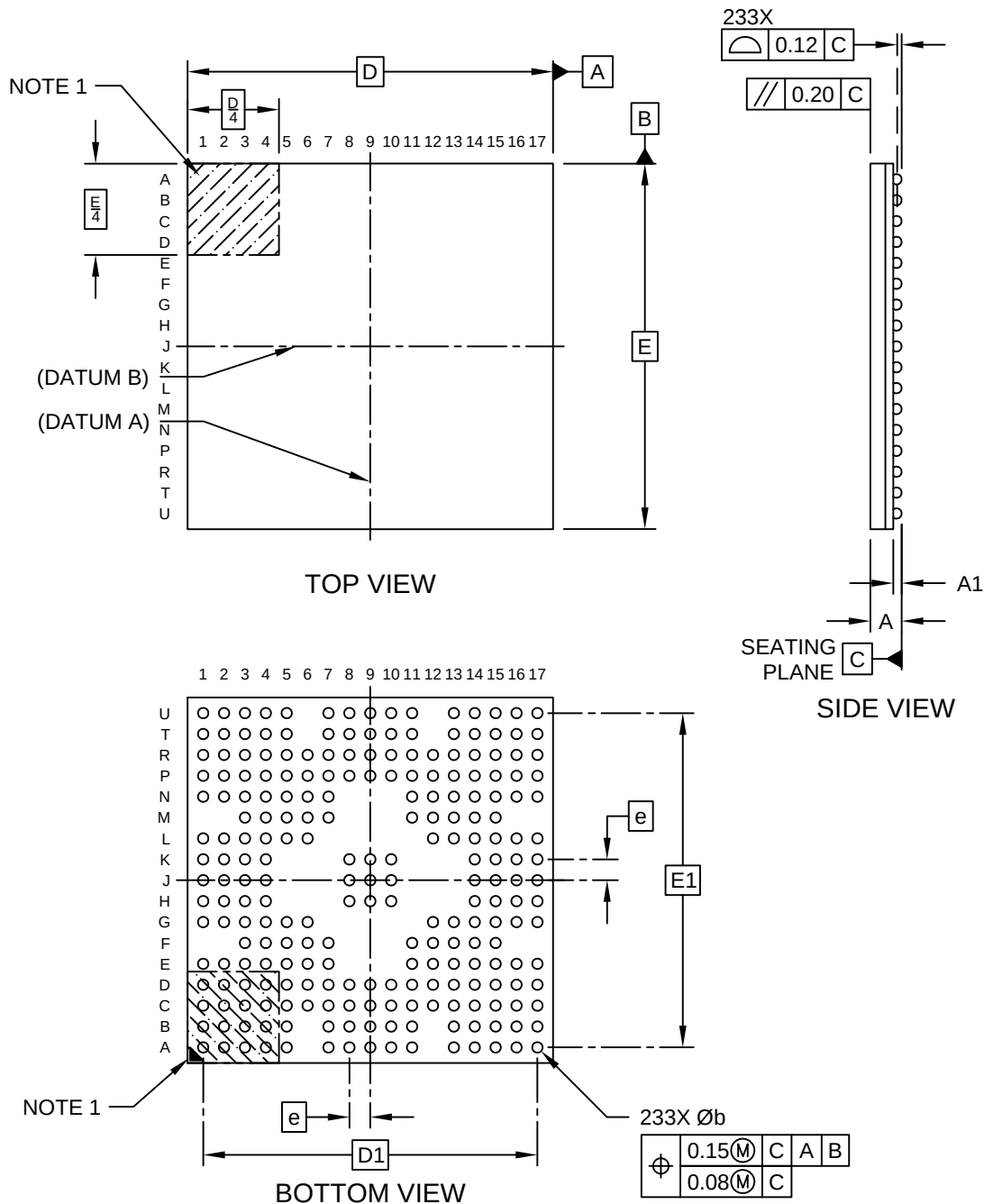


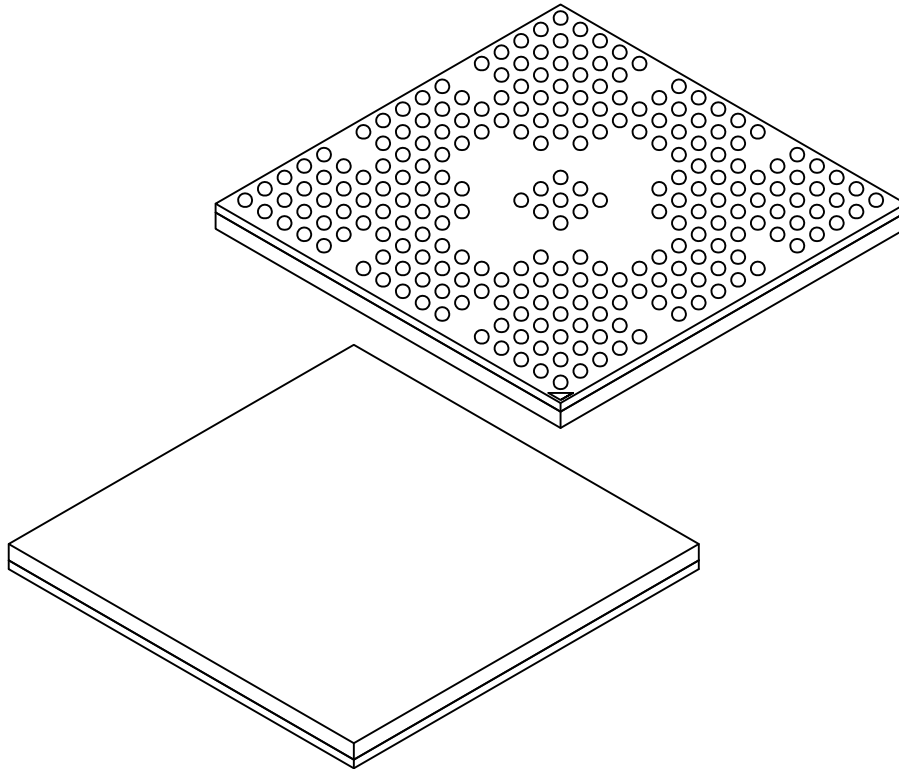
233-Ball Thin Fine Pitch Ball Grid Array (4FB) - 14x14 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



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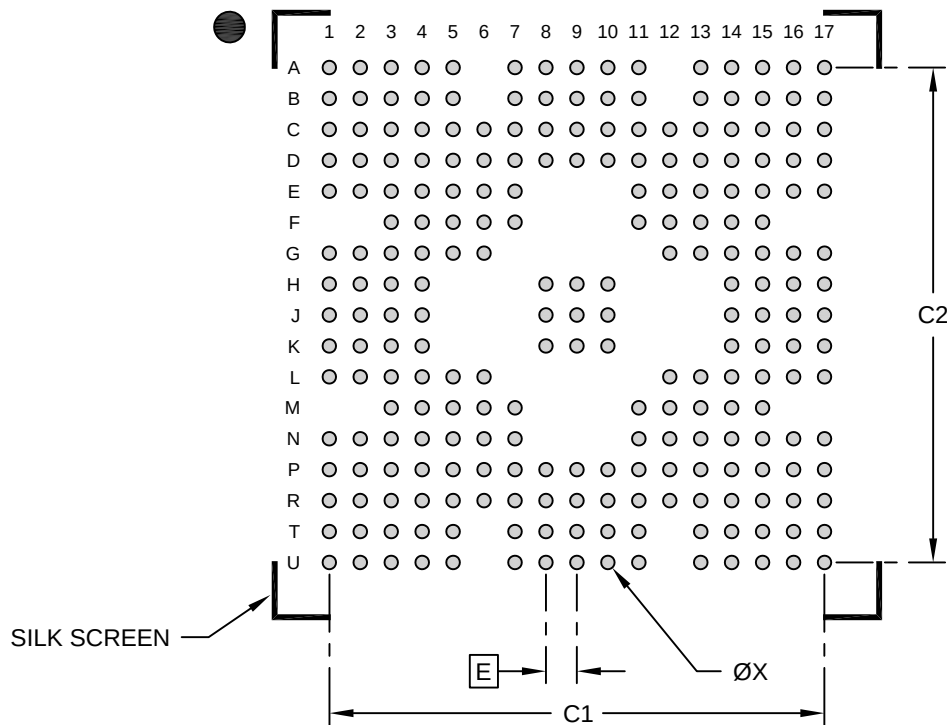
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	233		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.20
Standoff	A1	0.27	0.32	0.37
Overall Length	D	14.00 BSC		
Overall Ball Pitch	D1	12.80 BSC		
Overall Width	E	14.00 BSC		
Overall Ball Pitch	E1	12.80 BSC		
Terminal Width	b	0.38	0.40	0.48

Notes:

1. Terminal A1 visual index feature may vary, but must be located within the hatched area.
2. Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

233-Ball Thin Fine Pitch Ball Grid Array (4FB) - 14x14 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1		12.80	
Contact Pad Spacing	C2		12.80	
Contact Pad Width (Xnn)	X1			0.35

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23501 Rev A